SOIL EROSION AND CONSERVATION · NOTE GC03

Check Dams for Gully Control — Spacing, Design Rules, Brushwood, Loose-Rock and Gabion Check Dams

Course topics: SWCE4.5 L13, SWCE8.12 L8, SWCE-501 T12, SWCE8.15 L6

Gully control structures

- Mechanical structures across gullies or drainage lines that stabilise the grade, reduce velocity, trap sediment, let vegetation establish and protect the channel.
- **Temporary** structures — brush, poles, woven wire, loose rock, wooden planks — serve until vegetation establishes; **semi-permanent** ones use loose rock or dry stone; **permanent** ones are used when runoff is large and vegetation alone cannot cope. Supplement every mechanical structure with vegetation wherever possible.

Check dams

- The basic structure for controlling excessive gradients in drainage lines, gullies and ravines. They turn a steep slope into a series of steps — low risers with flat or gently sloping treads — reducing velocity and sediment-carrying capacity, stabilising the gradient, promoting vegetation, and increasing infiltration and groundwater recharge.
- **By function:** temporary, semi-permanent, permanent. **By structure:** live (vegetative) check dams, temporary, semi-permanent, gabion, and masonry or concrete check dams.
- **Components:** the **spillway** carries the design discharge; the **key** anchors the structure into the gully bed and sides; the **apron** dissipates the energy of the falling water.
- **Survey:** L-section (gradient and sudden drops), plan (drainage-line layout) and cross-sections (width and depth). Divide the drainage line into upper, middle and lower reaches.

Spacing

Ideally the top of the downstream dam is level with the bottom of the upstream dam. In practice, space them for a gentle, non-erosive grade of the deposited sediment (≤ 0.5%).

$$L = 100 \cdot H/(M - N)$$

L = horizontal distance between check dams; H = height of check dam; M = original bed slope (%); N = proposed bed slope after sedimentation (%)

Illustration (not in the notes)
H = 1 m, original slope M = 5%, design sediment slope N = 1%
L = 100 × 1/(5 - 1) = 25 m

Design rules

- Extend the structure into the bottom and sides of the channel so it cannot wash out; give the spillway enough capacity.

- Prefer low dams — the overfall of a temporary dam should be ≤ 75 cm, as greater heights cause high water pressure, leakage and scour.
- **Notch:** pass the peak runoff of a 5–10 year recurrence (rational formula), designed like a rectangular or trapezoidal weir and made as wide as possible so flow does not concentrate. **Freeboard:** 15 cm.
- **Overfall protection:** an apron or platform of brush, poles, loose rock or wooden planks downstream.
- **Life of temporary check dams:** 3–8 years — long enough for sediment to build up and vegetation to establish.

Live check dams

Vegetative barriers across a drainage line at the rill or early gully stage: two rows of grass 20 cm apart, with one row of shrubs on the downstream side — e.g. *Vitex*, *Agave*, *Arundo donax*.

Temporary check dams

- Low structures across small drainage lines or shallow gullies where velocity and sediment load are low, built of local material — loose stone, brushwood, logs, planks, wire mesh. They trap fine soil, slow runoff, improve moisture, help vegetation and control channel erosion; cheap, easy and unlikely to fail.
- **Limits:** height ≤ 1 m; overfall ≤ 75 cm; life 3–8 years.

Type	Use	Key dimensions
Woven-wire check dam	Moderate-slope gullies, small drainage areas; crescent shape with the open end downstream; spreads water, slows it and promotes silting	Offset from abutments ≈ 1/6 of drainage-line width (6 m wide → ≈ 1 m); posts 1.2 m apart and 60–90 cm deep; trench 15 cm deep; wire 15–30 cm above ground; apron 1.2 m long, extending 60 cm each side; earth backing 1 : 2
Single-row brushwood dam	Small drainage areas, gullies 1.2–2 m deep	Sides trimmed to 1 : 1 over 3–4.5 m; posts 10–15 cm thick, 75–90 cm deep, 60–90 cm apart; litter layer 15 cm
Double-row brushwood dam	As above, stronger	Rows 90 cm apart; posts 50 cm apart, 80–120 cm deep; litter 15 cm; brush 30 cm
Plank or slab dam	Moderately large drainage areas where timber is available	Posts 1 m deep, 1.2 m apart; trench 30 cm deep
Log and pole dam	Where timber is available	A good bond with the gully bed and sides is hard to get
Double-row pole rock-fill dam	Poles with rock fill	Needs less rock — economical and strong

Semi-permanent check dams

Loose rock or dry stone masonry for small to medium drainage areas in steep, broad gullies.

Loose-rock check dam	Specification
Construction	Flat stones, keyed together, voids minimised, no mortar; centre lower than the sides to form the spillway
Top width	50 cm
Side slope	0.5 : 1
Foundation depth	30–50 cm
Height above ground	50 cm – 1 m
Keying into banks	0–60 cm
Apron	Large flat rocks extending 1 m downstream

Gabion check dams

- Stone-filled wire-mesh structures across drainage lines and main drainage channels — flexible, porous, stable, economical and eco-friendly; they retain debris, accumulate soil and let vegetation establish.

Gabion item	Specification
Top width	1 m
Foundation depth	30–50 cm
Height	1–3 m
Keying into banks	30–60 cm
Wire	10 gauge (≈ 3.15 mm)
Mesh	15 × 15 cm to 20 × 20 cm
Typical box	3 m × 1 m × 1 m

1. Fabricate the mesh boxes.
2. Clear the site and excavate the foundation (30–50 cm) and bank keys (30–60 cm).
3. Place the cages and fill with stone — large stones at the bottom and sides, small stones in the voids.

Gabion with ferro-cement core	Specification
Purpose	An impervious core lets the dam retain water and resist runoff pressure
Ferro-cement thickness	2.5 cm
Concrete foundation	1 : 3 : 6, 15 cm thick
Core embedment	7.5 cm
Sand layer	15 cm
Keying	30–60 cm

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Check dam spacing formula is

(a) L = H/(M − N) (b) L = 100H/(M − N)
(c) L = 100(M − N)/H (d) L = H(M + N)

Q2. A 1 m check dam on a 4% gully designed for a 0.5% sediment slope needs spacing of about

(a) 4 m (b) 28.6 m
(c) 50 m (d) 100 m

Q3. Maximum overfall height of a temporary check dam is

(a) 30 cm (b) 75 cm
(c) 1.5 m (d) 3 m

Q4. Temporary check dams last about

(a) 1 year (b) 3–8 years
(c) 15 years (d) 30 years

Q5. Check-dam notches are designed for peak runoff of recurrence

(a) 1–2 yr (b) 5–10 yr
(c) 25 yr (d) 100 yr

Q6. The component that anchors a check dam into the bed and sides is the

(a) apron (b) spillway
(c) key (d) wing wall

Q7. Woven-wire check dams are offset from the abutments by about

(a) 1/2 (b) 1/4
(c) 1/6 (d) 1/10

Q8. Top width of a loose-rock check dam is

(a) 25 cm (b) 50 cm
(c) 1 m (d) 2 m

Q9. Typical gabion box size is

(a) 1 × 1 × 0.5 m (b) 2 × 1 × 0.5 m
(c) 3 × 1 × 1 m (d) 5 × 2 × 1 m

Q10. Ferro-cement core thickness in gabion check dams is

(a) 1 cm (b) 2.5 cm
(c) 7.5 cm (d) 15 cm

Answers

Q1 — (b) L = 100H/(M − N)
Q2 — (b) 28.6 m — *100 × 1/(4 − 0.5).*
Q3 — (b) 75 cm
Q4 — (b) 3–8 years
Q5 — (b) 5–10 yr
Q6 — (c) key
Q7 — (c) 1/6 — *of drainage-line width.*
Q8 — (b) 50 cm
Q9 — (c) 3 × 1 × 1 m
Q10 — (b) 2.5 cm